INTERNAL

PROJECT PHOENIX

Project Phoenix Architecture Review

Evaluation, decisions, and reference design

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Revision History

Version	Date	Changes	Author
0.1	2026-05-12	Initial draft.	A. Rivera

Executive Summary

This document summarises the architecture review for Project Phoenix, covering the requirements catalogue, candidate solutions, the weighted trade-off matrix, and the resulting design decisions. The reference architecture is validated against the in-house staging environment.

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SECTION 1

Context

1.1 Problem statement

Project Phoenix needs a structured way to evaluate competing implementation approaches against documented requirements before committing to a design.

> **INSIGHT — Core problem**
> Decisions have been made informally so far. We need a repeatable framework that produces an auditable record of why a given approach was chosen.

1.2 Environment

TARGET ENVIRONMENT

Platform	Linux containers on a managed Kubernetes cluster
Storage	Object store + relational database
Network	Internal VPC with controlled egress
Compliance	Internal audit baseline

1.3 Scope

The deliverables are:

1. A consolidated requirements catalogue.
2. A weighted trade-off matrix evaluating candidate solutions.
3. Architecture decisions captured as ADRs.
4. A validated reference design.

> **WARNING — Out of scope**
> This document does not cover roll-out planning or operational hand-over.

SECTION 2

Requirements

The requirements use category prefixes to group concerns. Colors are passed explicitly per requirement — there's no hard-coded taxonomy, so use whatever naming convention fits your project.

`FUN-001` Standards compliance — MUST

The system **shall** produce outputs compliant with the relevant industry standard, as profiled in the project specification.

RATIONALE	Interoperability with all current consumers requires conformance to the relevant industry standard.
VERIFICATION	Inspection of artefacts against the published specification.
SOURCE	Consumer projects; industry baseline.

`SEC-006` Cryptographic baseline — MUST

All cryptographic operations **shall** use parameters meeting the minimum strength defined by the corporate security policy.

RATIONALE	Mandated by the corporate security policy.
VERIFICATION	Cryptographic configuration review.
SOURCE	Corporate security policy.

`OPS-008` Daily throughput capacity — SHOULD

The system **should** handle at least 1000 requests per day.

RATIONALE	Sized from projected workload plus 50% headroom.
VERIFICATION	Sustained 24-hour load test at target throughput.
SOURCE	Aggregated demand estimate.

`GOV-011` Audit evidence trail — MUST

The system **shall** provide an evidence trail capturing all lifecycle, administrative, and configuration-change events.

RATIONALE	Audit-readiness requires immutable, attributable records.
VERIFICATION	Review audit data for completeness and traceability.
SOURCE	Corporate audit baseline.

SECTION 3

Trade-off matrix

The matrix scores each candidate from 1 to 5 per criterion, then computes a weighted total. The full scoring rationale is in the annexes.

Candidate scoring summary

Candidate	Functional	Security	Operations	Governance	Cost	Weighted
Option A	5	4	3	4	4	**4.00**
Option B	4	4	5	3	5	**4.15**
Option C	3	5	4	4	3	**3.80**
Option D	2	3	3	2	5	**2.90**

Scoring: 1 = inadequate, 3 = adequate with caveats, 5 = strong native support. Weighted total uses the criterion weights defined in §2.

SECTION 4

Architecture decisions

ADR-001 Adopt Option B as the reference design

PROPOSED

2026-05-12

CONTEXT

Option B scored highest on the weighted matrix, driven primarily by its operations and cost profile. It trades some governance maturity for significant operational savings.

DECISION

Option B becomes the reference design for Project Phoenix. The governance gap is addressed by adding a complementary tooling layer (see ADR-002).

CONSEQUENCES

Team must invest in operator training for Option B. Two extra weeks of schedule needed to integrate the governance tooling. Existing prototypes based on Option A are deprecated.

SECTION 5

Validation

The reference design is validated against the in-house staging environment.

TODO — Outstanding items

- Stress-test sustained throughput against the SHOULD target.
- Schedule architect review of ADR-001.
- Document the governance tooling integration (ADR-002 pending).

RISK — Open risk

No production-equivalent dataset is available for the validation run. Synthetic data may not exercise corner cases that real traffic would.